

DLA: X-linked disorders

GNET DLA_Before Mendelian Inheritance: X-linked inheritance

Two parts:

A. X-linked disorders

B. Unexpected phenomena in Mendelian disorders

Dr. Sharmila Upadhyia
Email: shupadhyia@sgu.edu

Required pre-reading: DLA: X-linked inheritance

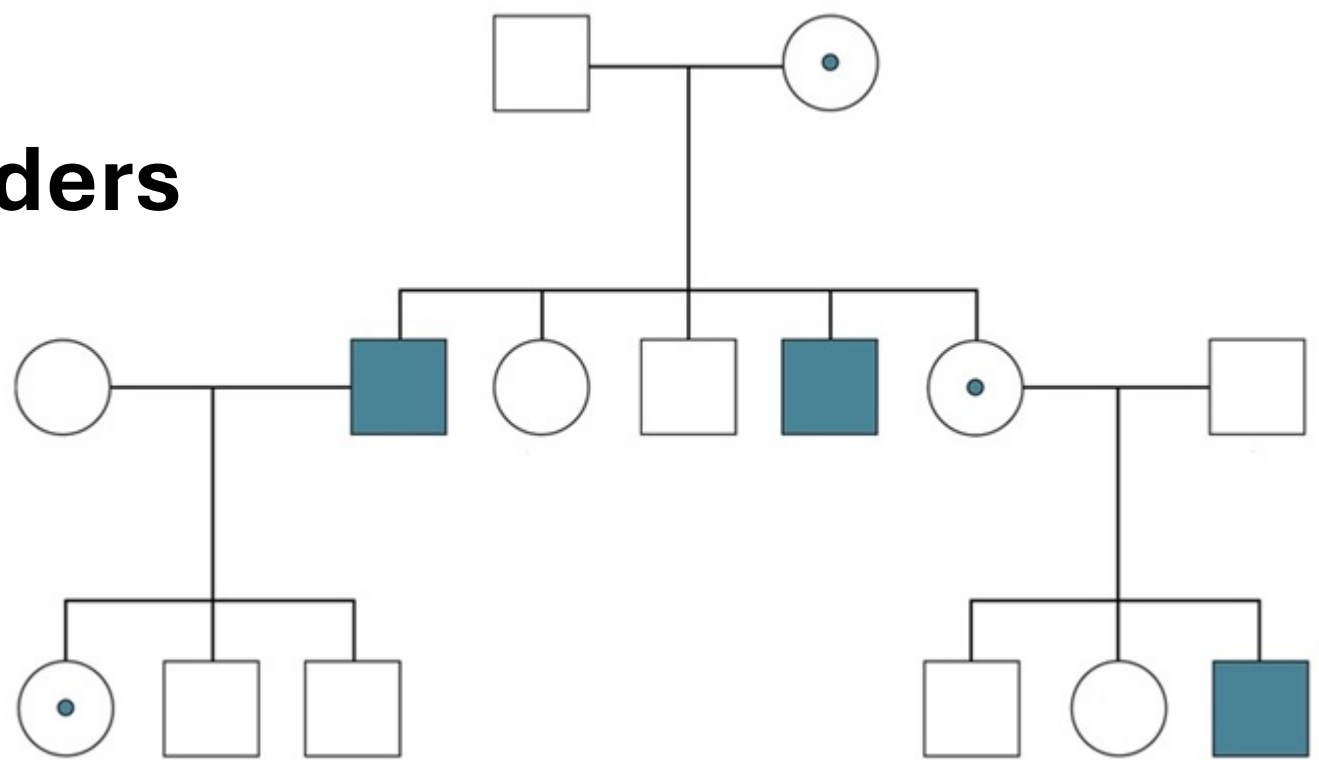
- DLA 12a: X-linkage examples
- DLA 12b: Unexpected findings in Mendelian inheritance (Shown with examples)

SOM.MK.III.BPM1.1.FTM.3.GNET.0037	Distinguish the mode of inheritance, identify the main clinical features, and interpret genetic principles illustrated by the following X-linked disorders: Hemophilia A and B, G6PD deficiency, Duchenne and Becker muscular dystrophy, red/green color blindness. Lesch Nyhan syndrome, X-SCID as examples of X-linked disorders
SOM.MK.III.BPM1.1.FTM.3.GNET.0038	Summarize and distinguish the characteristics of the following patterns of inheritance using pedigrees: autosomal dominant, autosomal recessive, X-linked dominant and X-linked recessive
SOM.MK.III.BPM1.1.FTM.3.GNET.0039	Demonstrate how disease liability is transmitted for X-linked diseases
SOM.MK.III.BPM1.1.FTM.3.GNET.0040	Propose how the basic risk calculations are performed for X-linked disorders
SOM.MK.III.BPM1.1.FTM.3.GNET.0041	Explain how a woman who is a carrier of an X-linked disorder might manifest symptoms of the condition (manifesting heterozygote) Skewed X-inactivation, non-random X-inactivation
SOM.MK.III.BPM1.1.FTM.3.GNET.0042	Discuss X-linked dominant disorders and traits
SOM.MK.III.BPM1.1.FTM.3.GNET.0043	Compare and contrast variable expressivity and incomplete penetrance; Give examples of each
SOM.MK.III.BPM1.1.FTM.3.GNET.0044	Describe pleiotropy
SOM.MK.III.BPM1.1.FTM.3.GNET.0045	Compare and contrast locus heterogeneity and allelic heterogeneity; gve examples
SOM.MK.III.BPM1.1.FTM.3.GNET.0046	Describe locus heterogeneity in terms of complementation groups
SOM.MK.III.BPM1.1.FTM.3.GNET.0047	Explain how allelic heterogeneity allows the formation of a compound heterozygote in autosomal recessive disorders

Objectives: Flipped class: X-linked inheritance

SOM.MK.III.BPM1.1.FTM.3.GNET.0048	Demonstrate how disease liability is transmitted for autosomal dominant diseases, autosomal recessive diseases and X-linked diseases
SOM.MK.III.BPM1.1.FTM.3.GNET.0049	Distinguish the different characteristics of recessive and dominant X-linked traits using pedigrees
SOM.MK.III.BPM1.1.FTM.3.GNET.0050	Identify obligate carriers in a pedigree depicting an X-linked trait
SOM.MK.III.BPM1.1.FTM.3.GNET.0051	Compare and contrast between X-linked, Y-linked, AR, AD, and mitochondrial traits.
SOM.MK.III.BPM1.1.FTM.3.GNET.0052	Discuss the concept of penetrance and variable expression and Differentiate between locus and allelic heterogeneity
SOM.MK.III.BPM1.1.FTM.3.GNET.0053	Calculate the recurrence risk based on penetrance for autosomal dominant and autosomal recessive disorders
SOM.MK.III.BPM1.1.FTM.3.GNET.0054	Define pleiotropy with examples. Be able to identify concepts if given a clinical scenario
SOM.MK.III.BPM1.1.FTM.3.GNET.0055	Recognize occurrence of new mutations as a cause of new genetic diseases in families. Be able to identify this concept from clinical scenarios and pedigrees
SOM.MK.III.BPM1.1.FTM.3.GNET.0056	Explain the phenomenon of germinal (germ-line) mosaicism
SOM.MK.III.BPM1.1.FTM.3.GNET.0057	Identify genetic disorders that have a delayed age of onset, and be able to identify this concept from clinical scenarios and pedigrees
SOM.MK.III.BPM1.1.FTM.3.GNET.0058	Explain the phenomenon of a manifesting heterozygote in X-linked recessive disorders, and be able to identify this concept from clinical scenarios and pedigrees

X-Linked Recessive disorders



- Preponderance of **affected males (Hemizygous for X-chromosome)**
- Skipping of generations
- **NO male-to-male transmission**
- Mothers of affected sons are **obligate carriers** of the mutant gene (I-1, II-3, III-1); Daughters of affected males are **obligate carriers**
- When mom is a carrier, the risk that her daughter is a carrier is 50% (II-3, III-5 in the pedigree)
- When a child is affected, look for the disorder in the **maternal male relatives (maternal uncle/grandfather/cousin)**
- Note that **recurrence risk changes with the sex of the parent carrying the mutant allele**

Recurrence risk for X-linked recessive disorder

Normal Father (XY) x Carrier mother (Xx*)

	Carrier mother	
Normal father	x*	X
X	x*X	XX
y	x*y	XY

Pathogenic allele

On average...

50% of daughters may be carriers; 50% of daughters may be normal

50% of sons may be affected; 50% of sons may be normal

Each conception event is independent
Risk of an affected child is $\frac{1}{4}$ (25%)

Recurrence risk for X-linked recessive disorder

Affected Father (x^*Y) x Normal mother (XX)

	Normal mother	
Affected father	X	X
Pathogenic allele → x^*	x^*X	x^*X
Y	XY	XY

All daughters of an affected father are obligate carriers of the mutant allele

A son **may not** inherit an X-linked mutation from his father

Risk of an affected child is NIL
No father-to-son transmission

Recurrence risk for X-linked recessive disorder

Affected Father (x^*Y) x Carrier mother (x^*X)

This is a relatively **rare occurrence** in population – usually seen for non-lethal X-linked traits (like red-green color blindness) where the carrier frequency is relatively high in the population, and males with the trait are not severely affected.

		Carrier mother	
Mutant allele →	Affected father	x^*	X
	x^*	x^*x^*	x^*X
	y	x^*y	XY

Each conception is independent event, **but on average**:

50% of daughters may be carriers; 50% of daughters may be affected (Homozygous)

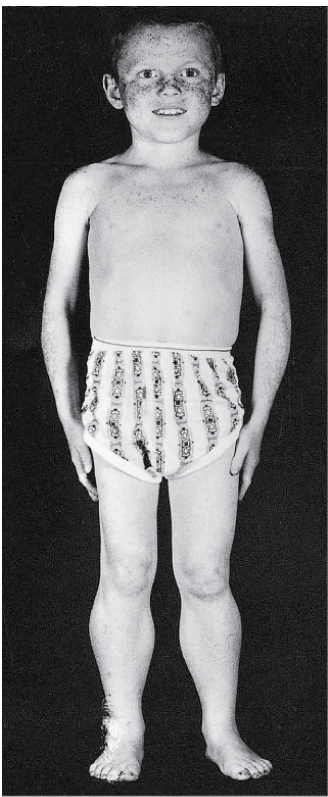
50% of sons may be affected; 50% of sons may be normal

X-Linked Recessive Disorders

- Dystrophin associated muscular dystrophy
 - Duchenne muscular dystrophy (severe)
 - Becker muscular dystrophy (milder form)
 - Both are due to mutations of the same gene (*DMD*, dystrophin)
- Hemophilia A and B result in bleeding disorders
- X-linked SCID (*IL2RG* gene mutation)
- Glucose 6-phosphate dehydrogenase (*G6PD*) deficiency
 - Hemolytic anemia on ingestion of primaquine, sulfa drugs
- Lesch-Nyhan syndrome [Hypoxanthine Guanine Phospho ribosyl transferase (*HGPRT*) deficiency]
 - Causes hyperuricemia, gout, & self mutilation
- Red-green color blindness/ deficiency (non-lethal)

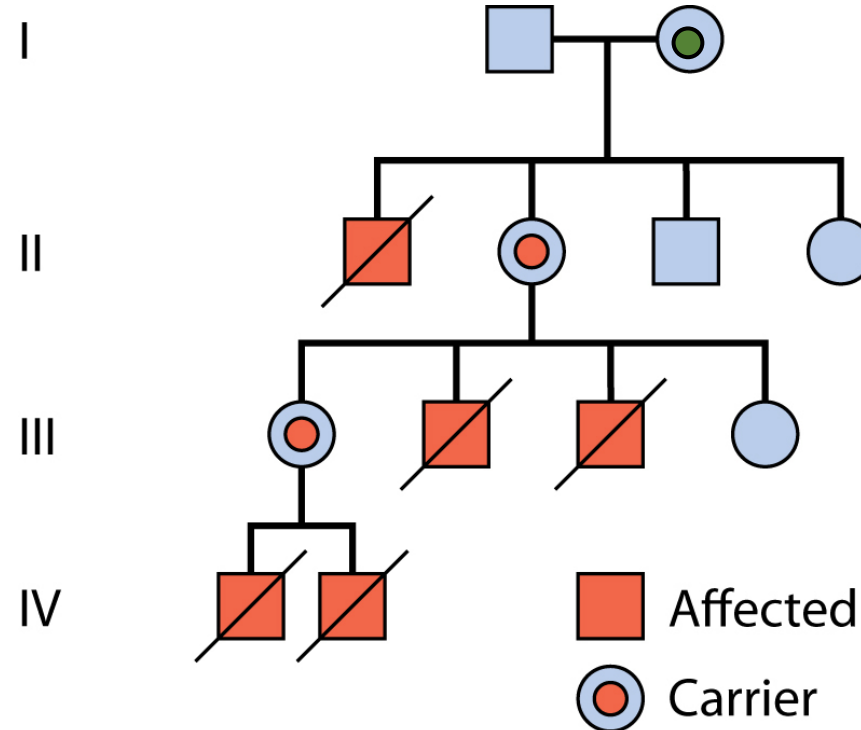


**X-linked enzyme
deficiency disorders**



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Duchenne Muscular Dystrophy (DMD)



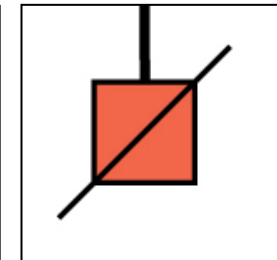
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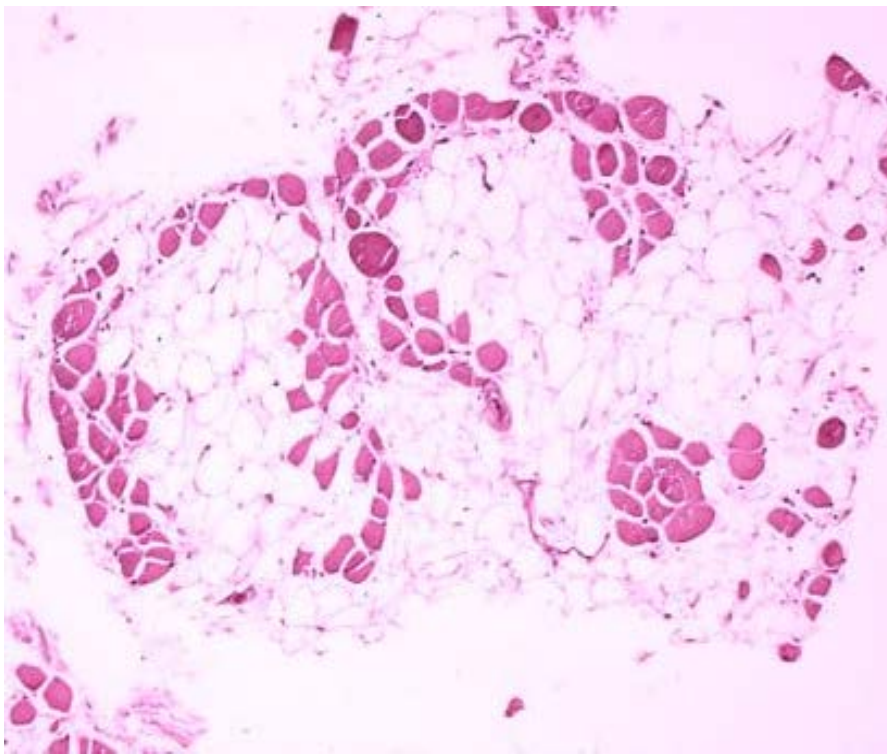
Family tree of Duchenne muscular dystrophy with the disorder being transmitted by carrier females and affecting males, who **do not survive** to transmit the disorder.

DMD: Note the enlarged calves (pseudohypertrophy of calf) and wasting of the thigh muscles.

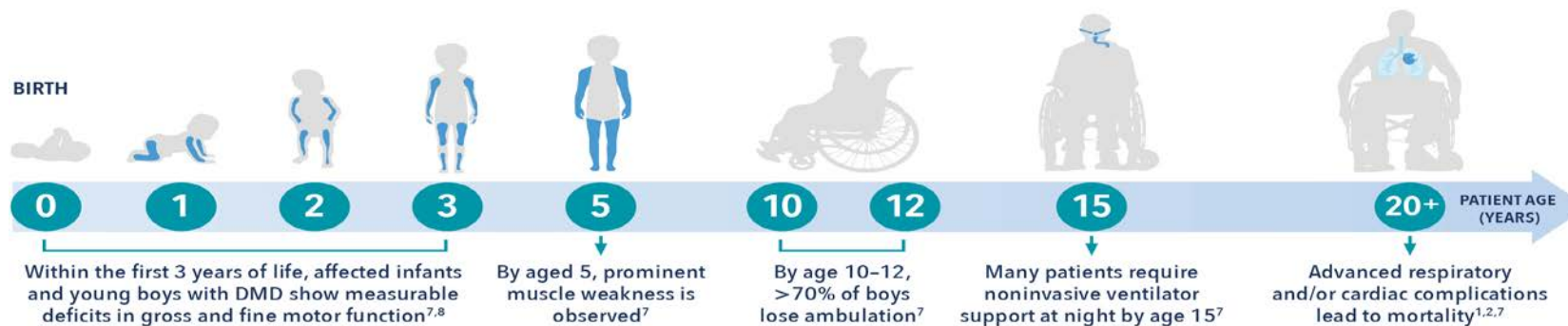
Calculate carrier risk of II-4 and III-4

Duchenne Muscular dystrophy tends to be lethal before the age of 30 (Males die), and they are so severely affected they usually don't have children – this means that DMD has a 'very low reproductive (genetic) fitness'





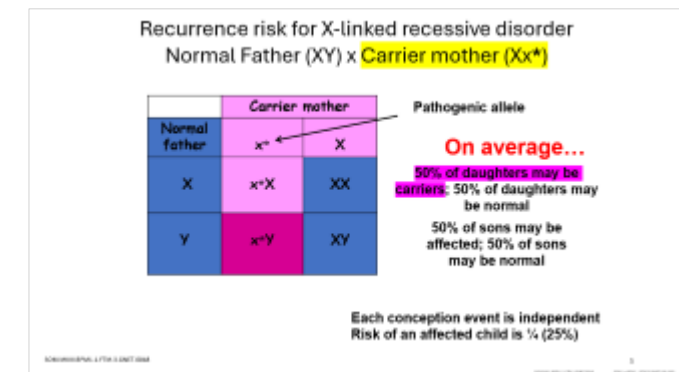
Histology of gastrocnemius muscle: Replacement of muscle fibres by adipose cells → Pseudohypertrophy of the calf muscle



Low reproductive (genetic) fitness

Duchenne muscular dystrophy

- DMD gene codes for dystrophin protein
- Daughters of carrier moms have a 50% carrier risk
- <https://www.youtube.com/watch?v=lolcG-a2yeY>



Becker muscular dystrophy

- Mutation is in the same gene (*DMD*) as in Duchenne muscular dystrophy
- Less severe mutations in *DMD* lead to Becker muscular dystrophy
 - Onset age
 - Amount of dystrophin
 - Reproductive fitness
- More severe mutations in *DMD* lead to Duchenne muscular dystrophy

Hemophilia A

- Inherited deficiency of clotting factor VIII → Increased tendency to bleed, after minor trauma
- A common mutation in factor VIII gene involve inversions of an intron sequence → Incorrect splicing → Loss of functional protein
- However, many other types of mutations of the **Factor VIII gene** result in Hemophilia A (**Allelic heterogeneity**)



Hemarthrosis in a patient with hemophilia



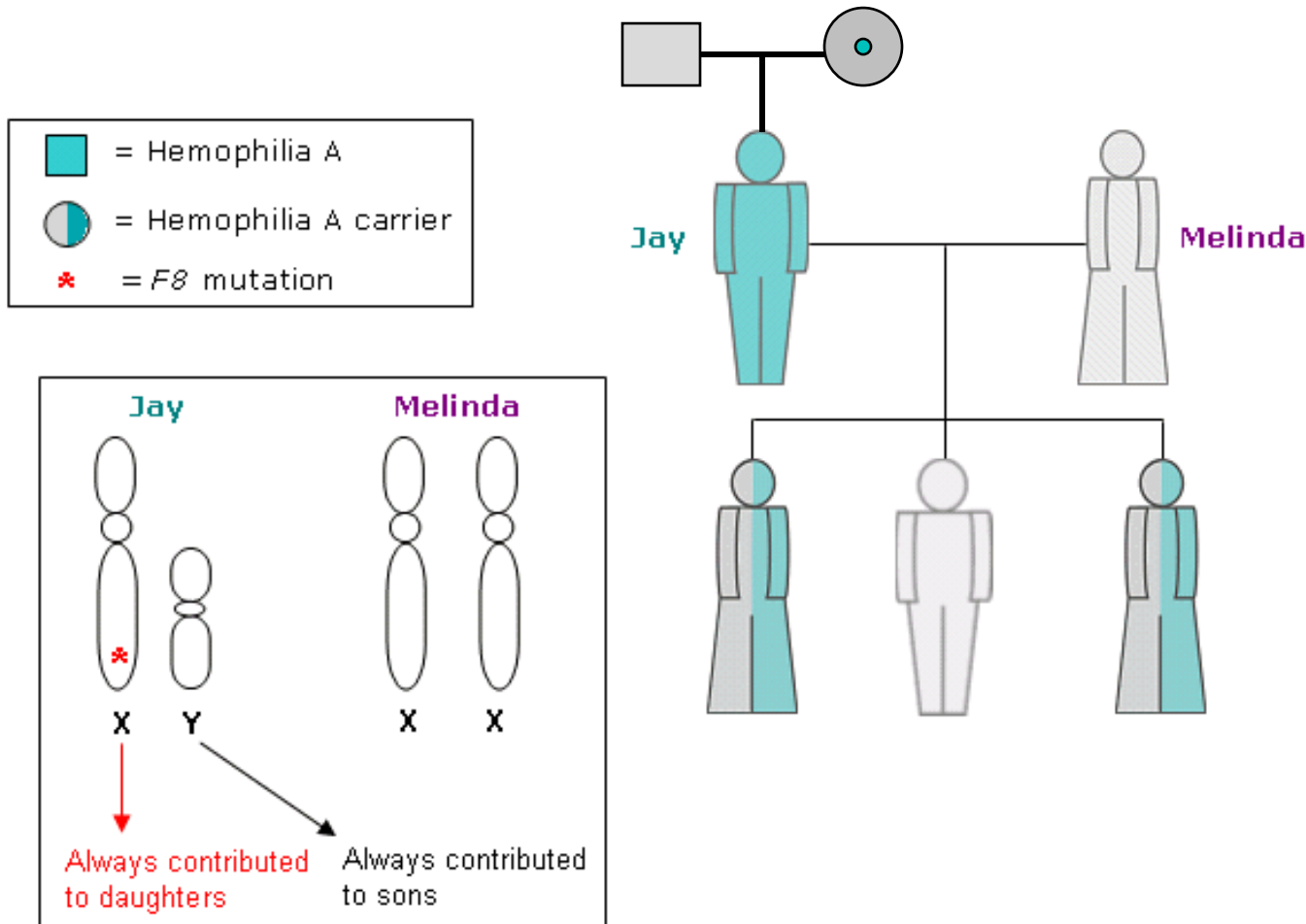
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Subcutaneous hematoma

X-linked SCID due to a defect in *IL2RG* gene

- Caused by a pathogenic variant in the γ -chain (gamma) of the receptor for several different interleukins (*IL2RG* gene)
 - Also called the γ c-cytokine receptor, since interleukins are cytokines
- If T-cells lack this receptor they cannot mature
- In turn, this results in a deficiency of normal B-cell function
 - Differentiate from autosomal recessive SCID
 - Locus heterogeneity

Obligate carriers in X-linked recessive disorders



Recurrence risk for X-linked recessive disorder
Affected Father (x^*Y) x Normal mother (XX)

	Normal mother	
	X	X
Affected father	X	X
Pathogenic allele x^*	x^*X	x^*X
Y	XY	XY

All daughters of an affected father are **obligate** carriers of the mutant allele

A son **may not** inherit an X-linked mutation from his father

Risk of an affected child is NIL
No father-to-son transmission

All daughters of a man affected with an X-linked disorder are heterozygous carriers of the mutation. Jay's mother is also most likely an obligate carrier of the mutation (other possibilities are a new mutation in the ovum or a germline mosaicism in the ova of Jay's mom)

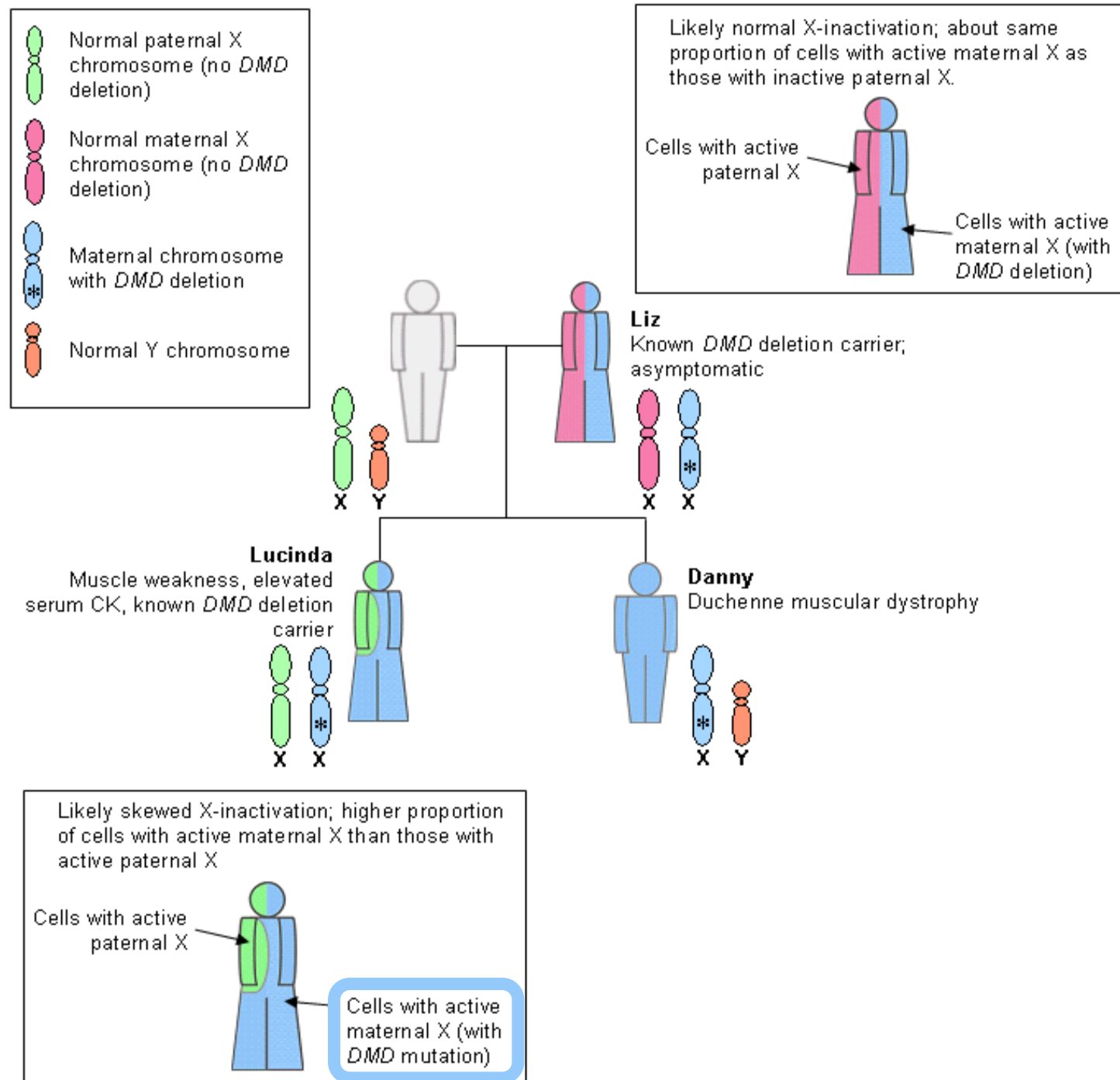
Manifesting heterozygote

- **A Woman with Postpartum Hemorrhage**
- A 37-year-old woman, experiences severe postpartum hemorrhage after delivering her first child. She also had difficulty with prolonged bleeding after her wisdom teeth were extracted.
- **A male first cousin** had bleeding problems and died from complications of **severe hemophilia A** and from complications of HIV at age of 25
 - He contracted HIV from transfusion during treatment
- Manifestations in a female are most commonly due to **skewed X-inactivation** (**asymmetric X-inactivation**)
- **If this asymmetric X-chromosome inactivation occurs, the carrier may have a factor VIII activity level below normal and thus experience bleeding problems.** These symptoms are generally mild compared to the bleeding problems of the affected male.

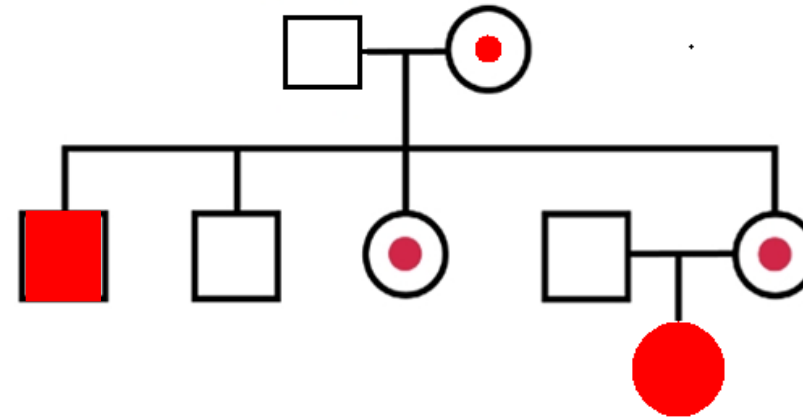
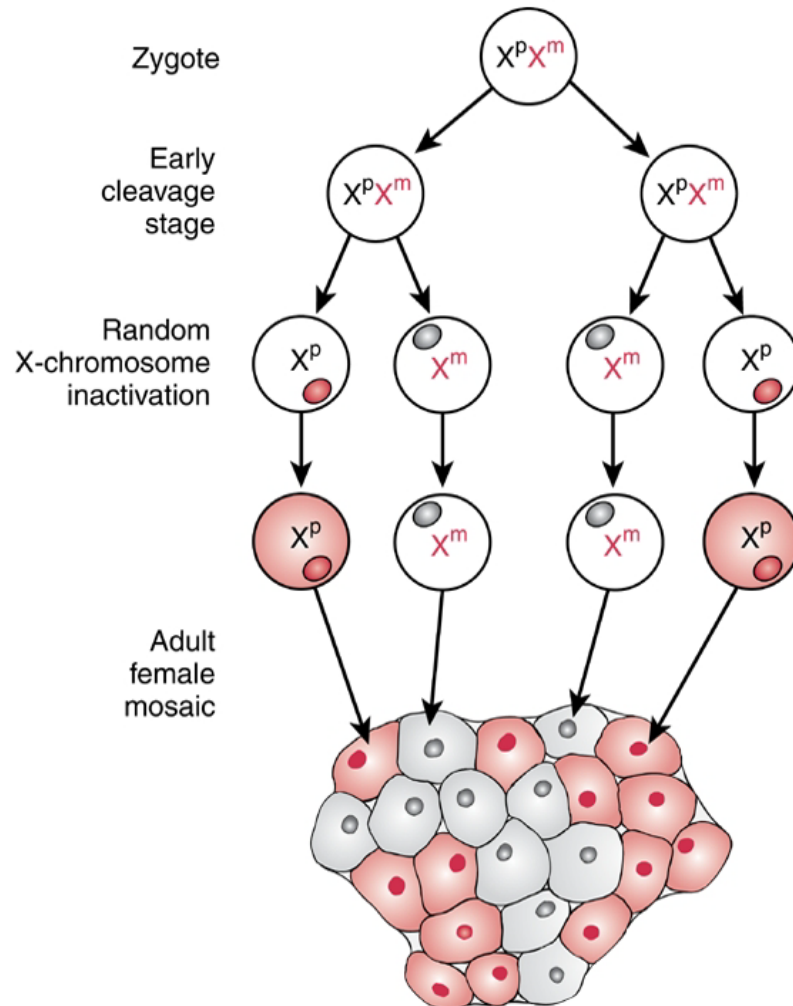
Manifesting heterozygote

- Women who are carriers of Duchenne muscular dystrophy alleles often have elevated creatine kinase levels
 - There are reports of muscular weakness in DMD carrier females. Examples of ***'manifesting heterozygotes'***
- This may be due to ***skewed X-inactivation (skewed lyonization)*** in these females
 - Note that X-inactivation occurs in somatic cells in females; Generally, about 50% of cells maternal X is active and 50% cells paternal X is active
 - The number of cells that contain the **active X chromosome with the mutation in DMD are large** compared to the cells that contain the active X with the functional DMD gene

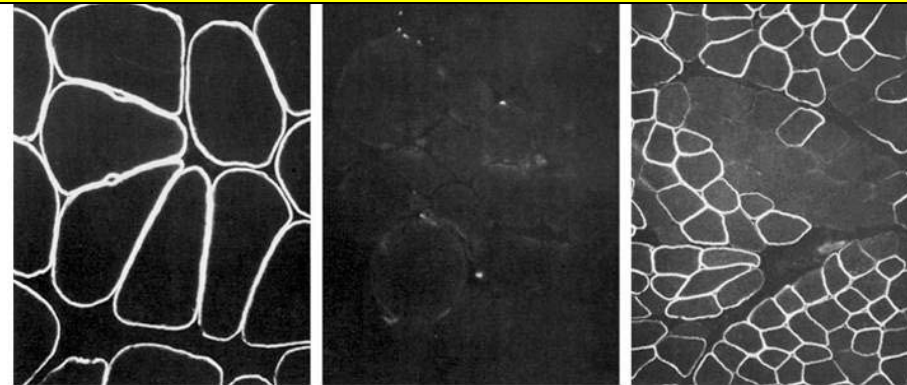
Skewed X-inactivation



Skewed X-Inactivation in DMD mutation carrier females who manifest the disorder



Areas of tissue can be healthy while other areas may be badly affected.



Red/green color deficiency

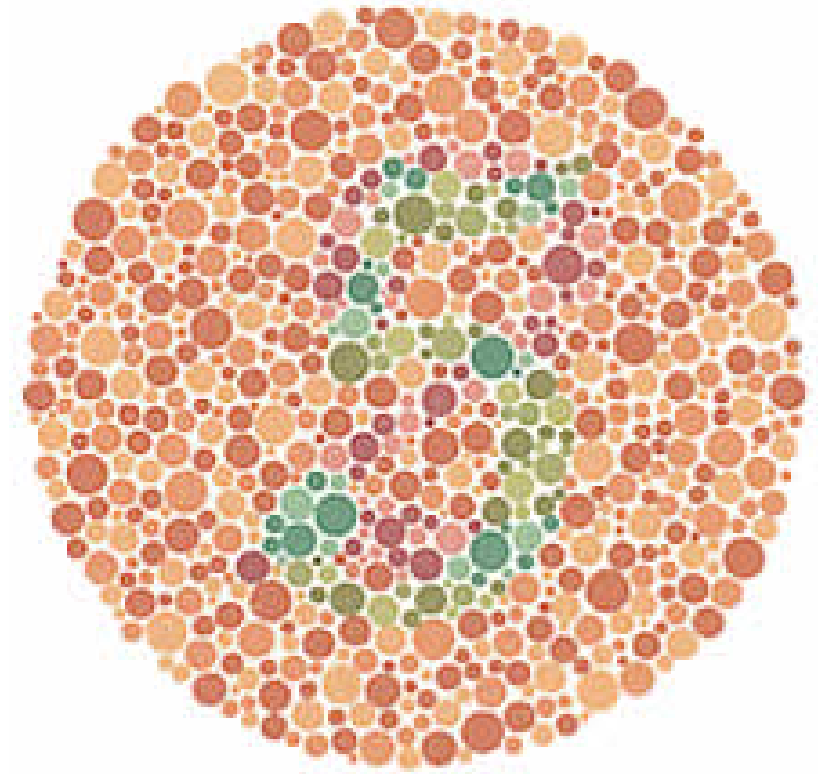
- **An example of Non-Lethal Sex-linked trait**
- Males are hemizygous for red/green color blindness. 8% of the male population is red/green color deficient
- Carrier females are prevalent
- Homozygous, color-blind females (1 in 150) exist but are much rarer than color blind males

Recurrence risk for X-linked recessive disorder
Affected Father (x^*Y) x Carrier mother (x^*X)

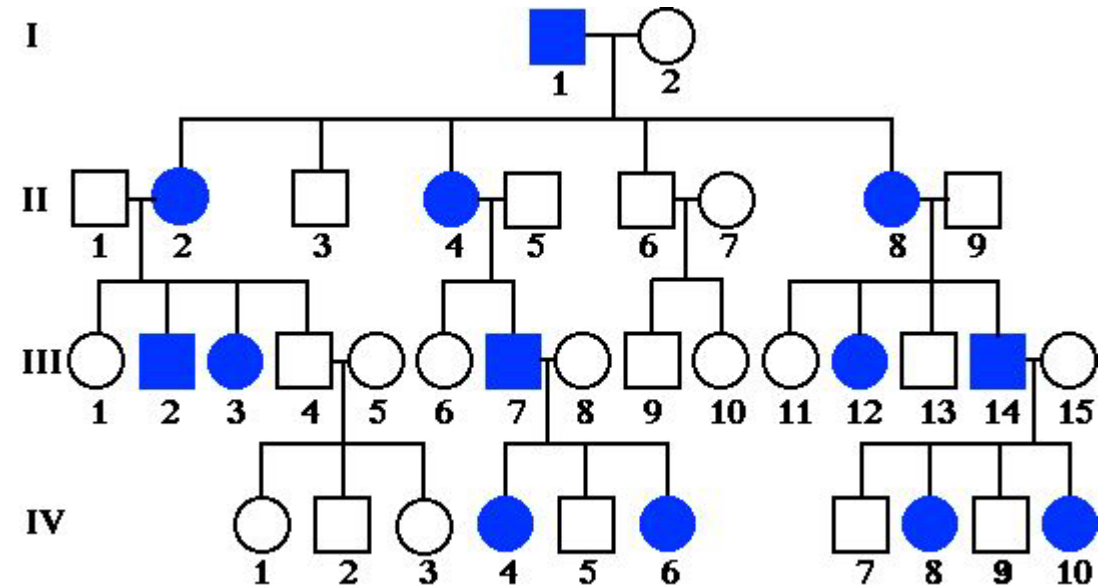
This is a relatively rare occurrence in population – usually seen for non-lethal X-linked traits (like red-green color blindness), where the carrier frequency is relatively high in the population, and males with the trait are not severely affected.

	Carrier mother	
	x^*	X
Affected father	x^*	X
Mutant allele	x^*	x^*X
	Y	x^*Y

Each conception is independent event, but on average:
50% of daughters may be carriers; 50% of daughters may be affected (Homozygous)
50% of sons may be affected;
50% of sons may be normal



X-linked dominant



- No skipping of generations;
- **No male-to-male transmission**
- Affected male → All his daughters affected
- Variable disease severity in females due to X-inactivation
- Some X-linked dominant disorders are lethal in male fetuses

X-linked dominant disorders

- Hereditary hypophosphatemic rickets
 - Patients have low serum phosphate levels
 - Do not respond to vitamin D
- Rett syndrome (lethal in males)
- Incontinentia pigmenti (lethal in males)

Recurrence risk for X-linked dominant disorder (or traits)

Affected Father (X^*Y) x Normal mother (XX)

		Normal mother	
Chromosome with mutant allele →	Affected father	X	X
	X^*	$X^* X$	$X^* X$
	y	xy	xy

All daughters would have the mutant allele and are affected

All sons are normal

Recurrence risk for X-linked dominant disorder

Normal Father (XY) x **Affected mother (XX*)**

Chromosome with
mutant allele

On average...

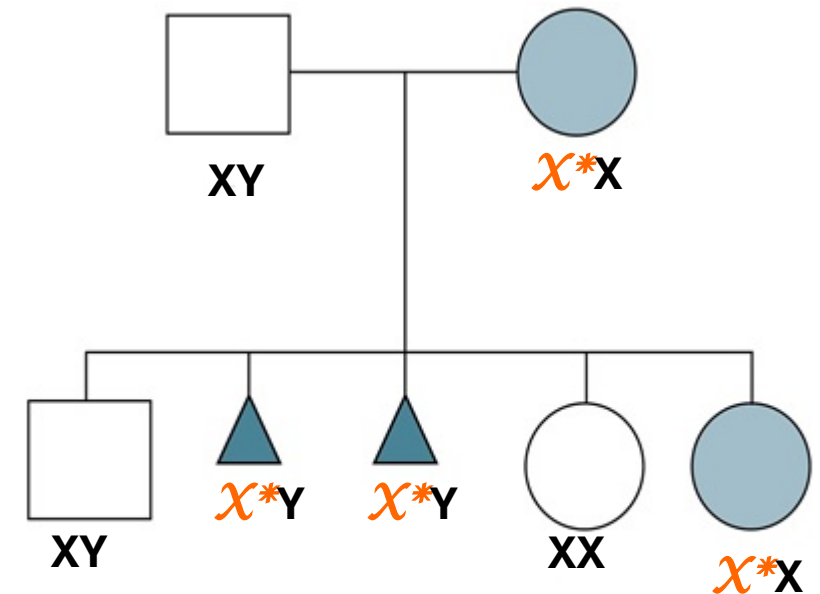
- 50% chance daughters will be affected
- 50% chance daughters may be normal

50% of sons may be affected and
50% of sons may be normal

	Affected mother	
Normal father	X*	X
X	X* X	XX
y	X*y	XY

Rett syndrome: X-linked disorder (lethal in males)

- Almost exclusively affects females
- Males with the mutant X-gene, usually die in utero (spontaneous abortion) or soon after birth
- Affected females: Autism-like features; wringing of hands; speech loss
- Rett syndrome: Clinical snapshot 3.2



Human Genetics and Genomics, Fourth Edition. Bruce R. Korf and Mira B. Irons.
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Figure 3.16 Pedigree illustrating X-linked dominant transmission with male lethality. Only females are affected. Males who inherit the mutant allele die in utero.

Incontinentia pigmenti: X-linked dominant (lethal in males)

- Males with the mutant allele die in utero (male-lethal)
- Females
 - Rashes, blisters → Hyperpigmentation patches ('Marble cake' skin)
 - Intellectual and learning disability (in some patients)
 - Retinal detachment (in some patients)
- Variable expressivity in females: Due to X-inactivation (Female mosaics)
- Patchy, darker pigmentation where the normal X has been inactivated (or mutant X is active)
- *Areas of normal pigmentation indicate areas where normal X is active*



Summary: X-linked disorders

- Calculation of recurrence risks and recognition of pedigrees
- Calculation of carrier risks in X-linked recessive disorders
- Genetic principles
 - Manifesting heterozygote (skewed X-inactivation)
 - Hemizygous
 - Reproductive fitness
 - Allelic and locus heterogeneity
- Clinical features and genetic mechanisms of the disorders listed

X-Linked Recessive Disorders

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 - Duchenne muscular dystrophy (severe)
 - Becker muscular dystrophy (milder form)
 - Both are due to mutations of the same gene (DMD, dystrophin)
- Hemophilia A and B result in bleeding disorders
- X-linked SCID (IL2RG gene mutation)
- Glucose 6-phosphate dehydrogenase (G6PD) deficiency
 - Hemolytic anemia on ingestion of primaquine, sulfa drugs
- Lesch-Nyhan syndrome [Hypoxanthine Guanine Phospho ribosyl transferase (HGPRT) deficiency]
 - Causes hyperuricemia, gout, & self mutilation
- Red-green color blindness/ deficiency (non-lethal)

X-linked enzyme deficiency disorders

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shupadhya@sgu.edu if
you have any questions

X-linked dominant disorders

- Hereditary hypophosphatemic rickets
 - Patients have low serum phosphate levels
 - Do not respond to vitamin D
- Rett syndrome (lethal in males)
- Incontinentia pigmenti (lethal in males)